ALGEBRA 2 · FUNCTIONS & TRANSFORMATIONS

# Transformation Lab

Explore & Reflect

Name _____ Date _____ Period _____

Open **Transformation Lab** on Mr. K's Classroom: Algebra 2 → Functions & Transformations → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

## 1 Predict before you touch the tool

1. Where is the vertex of $f(x) = |x - 2| + 3$?

2. What does a negative value of a do to a graph?

## 2 Explore use the tool and record what you see

**Transform the parent**

Set each function and record what happened.

| Function | Parent | Transformations | Key point |
|---|---|---|---|
| $g(x) = (x + 4)^2 - 1$ | $x^2$ | | |
| $g(x) = -\|x\| + 2$ | $\|x\|$ | | |
| $g(x) = \sqrt{(x - 3)}$ | $\sqrt{x}$ | | |
| $g(x) = \sqrt{(-x)}$ | $\sqrt{x}$ | | |
| $g(x) = 2(x - 1)^3$ | $x^3$ | | |
| $g(x) = \frac{1}{(x - 2)} - 1$ | $\frac{1}{x}$ | | |
| $g(x) = 2^{x+1}$ | $2^x$ | | |

## 3 Write the rule fill in each blank from what you observed

1. $f(x) + k$ shifts the graph _____. $f(x - h)$ shifts it _____.

2. $-f(x)$ reflects the graph over the _____. $f(-x)$ reflects it over the _____.

3. $a \cdot f(x)$ with $|a| > 1$ is a vertical _____. $f(bx)$ with $|b| > 1$ is a horizontal _____.

4. Changes inside the function act on the _____ values. Changes outside act on the _____ values.

## 4 Check no tool: show your work

1. Describe $g(x) = -3\sqrt{(x + 1)} - 2$.

2. Write $|x|$ shifted right 5 and down 4.

3. Where does the center of $x^3$ move in $g(x) = (x - 2)^3 + 5$?

4. Move $\frac{1}{x}$ so its asymptotes are $x = -3$ and $y = 4$.

## 5 Reflect

Why does $f(x - 2)$ move the graph right instead of left?